STRUCTURAL CONCRETE AND REINFORCING

This specification has been revised as indicated below and described in the revision record on the following page. Please destroy all previous revisions.

Revision No.	Date	Originator's Name & Initials	Reviewed/Checked By Name & Initials	Pages

APPROVALS	**SIGNATURES**	**DATE**
Lead Engineer		
Project Manager:		
Client Approval:		

ISSUED FOR : ☐ Construction ☐ Other ____________

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Record of Revisions

Revision No.	Date	Description

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Section

STRUCTURAL CONCRETE AND REINFORCING

Note!!! Notes, including this one, are to be reconciled and removed when this is issued as a Project Specification. This specification prescribes a uniform format. The Lead Structural Engineer has the responsibility to modify, add, or delete, the following standard provisions in order to meet project specific requirements.

1.0 GENERAL

1.1 Summary

A. Scope of Specification

This specification prescribes requirements for cast-in-place concrete construction.

1.2 References

The publications listed below form part of this specification to the extent referenced. If there is a discrepancy between the references and this specification, then this specification shall govern.

Note!!! Engineer to verify current dates of the referenced documents before issuing the specification.

A. ACI (American Concrete Institute)

1. ACI 117-90 Specification for Tolerances for Concrete Construction and Materials
2. ACI 211.1-91 Standard Practice for Selecting Proportions for Normal, Heavyweight, and Mass Concrete
3. ACI 211.2-98 Standard Practice for Selecting Proportions for Structural Lightweight Concrete
4. ACI 301-96 Specification for Structural Concrete
5. ACI 302.1R-96 Guide for Concrete Floor and Slab Construction
6. ACI 304R-89 Guide for Measuring, Mixing, Transporting, and Placing Concrete
7. ACI 304.2R-96 Placing Concrete by Pumping Methods
8. ACI 305R-91 Hot Weather Concreting
9. ACI 306.1-90 Specification for Cold Weather Concreting
10. ACI 306R-88 Cold Weather Concreting
11. ACI 308-92 Standard Practice for Curing Concrete

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12. ACI 347R-94 Guide to Formwork for Concrete

13. ACI 504R-90 Guide to Sealing Joints in Concrete Structures

14. ACI SP-66 (94) ACI Detailing Manual

B. **ASTM (American Society for Testing and Materials)**

Note!!! ASTMs noted in Italic shall only be included if used in Section 2.2B.

1. ASTM A82-97a Specification for Steel Wire, Plain, for Concrete Reinforcement

2. ASTM A615-96a Specification for Deformed and Plain Billet-Steel Bars for Concrete Reinforcement

3. ASTM C94/C94M-98c Specification for Ready-Mix Concrete

4. ASTM C150-97a Specification for Portland Cement

5. ASTM C171-97a Specification for Sheet Materials for Curing Concrete

6. ASTM C618-98 Specification for Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use as a Mineral Admixture in Concrete

7. ASTM C989-97b Specification for Ground Granulated Blast-Furnace Slag for Use in Concrete and Mortars

8. ASTM C1059-91 Specification for Latex Agents for Bonding Fresh to Hardened Concrete

9. ASTM C1240-98 Specification for Silica Fume for Use as a Mineral Admixture in Hydraulic-Cement Concrete, Mortar, and Grout

10. ASTM D1056-98 Specification for Flexible Cellular Materials - Sponge or Expanded Rubber

11. ASTM D2103-97 Specification for Polyethylene Film and Sheeting

C. **NRMCA (National Ready Mixed Concrete Association)**

1. Certification of Ready Mixed Concrete Production Facilities, Seventh Edition, April 1999

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D. OSHA (Occupational Safety and Hazards Administration)

1. OSHA Construction Industry Standards: Title 29, Code of Federal Regulations, Part 1926, Safety and Health Regulations for Construction, Feb. 1996

1.3 ACI 301-96

A. Concrete construction shall conform to the requirements of ACI 301, Standard Specification for Structural Concrete, except as modified or supplemented by this specification or the design drawings. The reference standards listed in ACI 301, Section 1.4, are declared to be a part of this specification the same as if fully set forth herein.

Note!!! A number of provisions from the previous version of this specification were removed because they only served to repeat what is in ACI 301. Before inserting additional requirements, check ACI 301 for coverage.

B. This specification provides modifications, clarifications, and extensions and does not repeat the provisions of ACI 301. For provisions not mentioned herein, refer to ACI 301.

Note!!! The following listed documents are handled this way because they are all written in non-mandatory language and are not intended to be used as project specification requirements.

C. In meeting the requirements of this specification and ACI 301, the following documents shall be used as acceptable practice:

1. ACI 211.1, Standard Practice for Selecting Proportions for Normal, Heavyweight, and Mass Concrete

2. ACI 211.2, Standard Practice for Selecting Proportions for Structural Lightweight Concrete

3. ACI 302.1R, Guide for Concrete Floor and Slab Construction

4. ACI 304R, Guide for Measuring, Mixing, Transporting, and Placing Concrete

5. ACI 304.2R, Placing Concrete by Pumping Methods

6. ACI 305R, Hot Weather Concreting

7. ACI 306R, Cold Weather Concreting

8. ACI 308, Standard Practice for Curing Concrete

9. ACI 347R, Guide to Formwork for Concrete

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10. ACI 504R, Guide to Sealing Joints in Concrete Structures

1.4 Submittals

For submittals, refer to the Purchase Order/Contract.

Note!!! For reference, Attachment 01 of this specification includes a list of supplier data submittal requirements, which should be included in the PO/Contract. Attachment 01 and this note should be deleted from this specification before it is issued for a project. Alternately, the list of submittals may be inserted here or left as an attachment to this specification.

1.5 Quality Assurance

A. The Concrete Contractor shall have a written Quality Control Program and Inspection Procedures document that shall provide details of how compliance with the requirements of this specification and the shop and placement drawings shall be achieved.

B. Fluor Daniel reserves the right to make inspections at any time at the source of supply of materials, at the place of preparation of materials, at the mixing plant if ready-mixed concrete is used, and during execution of all concrete work.

C. Sets of four cylinders shall be molded and cured. One cylinder shall be tested at seven days for information and two cylinders at 28 days. The fourth cylinder shall be used as a spare and shall be tested as specified by the Fluor Daniel Construction Manager.

D. Ready mixed concrete supplier shall hold a current National Ready Mixed Concrete Association's Certificate of Conformance for Concrete Production Facilities. Alternatively, certification from an independent testing agent stating conformance with NRMCA "Certification of Ready Mixed Concrete Production Facilities" is acceptable.

Note!!! ACI does not explicitly define "Massive Concrete". For structures deemed to be defined as "Massive" where the requirements of ACI 301 Chapter 8 are to be invoked, shall either be noted on the design drawings or a definition similar to the following shall be added here: Massive Concrete shall be defined as any concrete element with a least dimension of 3 feet and a total volume exceeding 8 cubic yards.

2.0 PRODUCTS

2.1 General

A. Products not meeting the criteria given in this specification shall not be used unless authorized in writing by the Fluor Daniel Lead Structural Engineer.

B. Products shall meet applicable local Volatile Organic Compound regulations.

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2.2 Materials

A. Cement

Note!!! If other than ASTM C150 Type I cement is required, it shall be specified here as follows:

1. Cement: ASTM C150 Type [::]. Use only one brand of cement.

B. Mineral Additives

Note!!! Mineral additives may be required (or optional) to improve durability, lower the heat of hydration, or for economical reasons. If the mineral additive percentage desired or required is different than as shown in ACI 301 Section 4.2.2.8b, then this shall be stated. If any mineral additives are specified, the applicable ASTM shall be added to Section 1.2B. Where mineral additives are deemed required or may be considered optional by the concrete contractor, they shall be specified and meet one of the following ASTM specifications:

1. Fly ash: ASTM C618, Class F, unless Class C is specified on the design drawings

2. Ground granulated blast-furnace slag: ASTM C989, Grade 100 or 120

3. Silica Fume: ASTM C1240 (and optional chemical or physical requirements, if required).

C. Aggregates

1. A single source shall be used for the supply of aggregate.

D. Water

1. Water that is not potable shall be tested for suitability in accordance with ASTM C94 for questionable water supplies.

E. Admixtures

1. All admixtures require the Fluor Daniel Lead Structural Engineer's authorization for use.

2. Admixtures from only one supplier shall be used.

3. All admixtures shall be added at the batch plant.

4. Calcium Chloride shall not be used.

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F. Accessories

1. Bonding compounds: nonrewettable, polyvinyl acetate type: ASTM C1059, type II.

2. Epoxy adhesive compounds: two component, 100 percent solids, 100 percent reactive compound suitable for use on dry or damp surfaces.

3. Repair toppings: self-leveling, polymer modified high strength topping.

4. Vapor barriers: 6 mil clear polyethylene film, or a fabric reinforced plastic film of a type recommended for below grade applications.

5. Patching mortars: free-flowing polymer-modified cementitious coating.

G. Curing Compounds

1. Plastic coverings: ASTM D2103 minimum 6 mils thick clear polyethylene film.

2. Absorptive mats: ASTM C171, made of burlap-polyethylene, weigh a minimum of 8 ounces per square yard and bonded to prevent separation during use.

H. Joints

1. Joint primer: two components penetrating liquid resinous primer for use with urethane and epoxy sealants.

2. Backup material: ASTM D1056, round closed cell foam rod, sized 30 to 50 percent larger than the joint width.

3. Waterstops: dumbbell or centerbulb type extruded PVC, minimum width of 6 inches.

I. Reinforcing

1. Reinforcing bars: deformed billet steel, ASTM A615, grade 60.

Note!!! ASTM A706 may be required for high seismic zones or for weldability.

Note!!! The following statement should be modified if galvanized or epoxy coated reinforcing is needed. The appropriate ASTM reference specifications should not be included here because they are included in ACI 301.

2. All reinforcing shall be uncoated.

3. Tie wire: black annealed wire, 16 gage minimum.

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2.3 Design Criteria

A. Reinforcing Detailing and Fabrication

1. Placing drawings and bending schedules showing the number, grade, size, length, mark, location, and bending diagrams for reinforcing steel shall be prepared in accordance with the ACI Detailing Manual.

2. Splices shall be only as shown on the drawings.

3. Each structure or foundation will have a different identification number on the design drawing. Fabrication drawings shall indicate the related PO number, identification number, and design drawing number.

4. Tag reinforcing using weather resistant metal tags.

5. Tag each bundle of fabricated bars showing drawing number, release number, mark number, bar quantity, and bar size.

6. Tag each bundle of stock length straight bars showing bar quantity, bar size, and bar length.

B. Concrete Mix Criteria

1. The minimum 28-day compressive strength f 'c shall be as follows unless noted otherwise on the design drawings:

   a. Foundations: 4,000 psi
   b. Paving: 3,000 psi
   c. Structures: 4,000 psi
   d. Underground electrical ducts: 2,000 psi
   e. Fireproofing: 3,000 psi
   f. Manholes, drain boxes, ditch lining, end walls, curbs: 3,000 psi

2. If high early strength concrete is specified, compressive strength shall be 7-day strength.

3. The maximum water-cementitious materials ratio shall be 0.40 for water retaining (hydraulic) structures and 0.45 for all other structures.

Note!!! In harsh environmental conditions, a w/c ratio equal to or less than 0.40 may be required for all exposed concrete, especially in hot and arid areas with high concentrations of sulfates and chlorides. In addition, these areas may also require the use of High Range Water Reducing Agents (superplasticizers). Note also that additional ASTM tests may be required for evaluating durability such as Rapid Chloride Permeability Tests, Water Penetration, Porosity, 90 Day

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Ponding Chloride Permeability, Water Absorption and tests for Total Chlorides and Sulfates.

4. Unless noted on the design drawings, the maximum water soluble chloride ion concentrations in hardened concrete from all sources shall be as indicated in ACI 301 Table 4.2.2.6 for "[insert type of member here]".

5. The maximum aggregate size shall be 3/4 inches for electrical ducts, slabs and walls less than 8 inches thick, elevated slabs, lightweight concrete, and water retaining structures. For fireproofing, the maximum aggregate size shall be 3/8 inch. For all other items, the maximum aggregate size shall be 1-1/2 inches.

6. Concrete envelopes for underground electrical ducts shall be colored red by adding 15 pounds of red oxide powder per cubic yard of concrete.

7. Concrete envelopes for underground instrument air ducts shall be colored yellow by adding 15 pounds of yellow oxide powder per cubic yard of concrete.

Note!!! Color additive quantities should be verified with the local concrete supplier. Lessor amounts may be justified.

3.0 EXECUTION

3.1 General

A. All work shall be in accordance with OSHA safety requirements.

B. The Concrete Contractor shall obtain inspection and authorization from the Fluor Daniel Construction Manager before placing concrete.

C. Form release agents shall not be applied where concrete surfaces will receive special finishes or where the agent affects applied coverings. Alternately, soak inside surfaces of untreated forms with clean water and keep surfaces moist before placing concrete.

D. Place slabs in alternating strips.

E. Where the minimum temperature criteria of ACI 301, Section 4.2.2.7 applies, ACI 306.1 shall be followed.

F. Where the combination of temperature, humidity, and wind velocity per ACI 305 Figure 2.1.5 are expected to cause a rate of evaporation equal to or greater than 0.2 lb/sq. ft/hr, the recommendations of ACI 305 shall be followed.

G. Earth cut forms for footings may be used with the approval of the Fluor Daniel Construction Manager.

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3.2 Reinforcing and Embedments

A. The Contractor shall protect bolt threads against damage and concrete and shall cap or plug anchor bolt sleeves to keep out water, concrete, and debris.

B. Tack welding of anchor bolts, reinforcing steel, and embedments is not permitted.

C. Do not insert dowels into fresh concrete.

D. Continue reinforcing through construction joints. Interrupt reinforcing at isolation joints and provide and install dowels as detailed on the drawings.

E. Splice reinforcing bars only as shown on the design drawings. Welded splices are not allowed unless detailed and approved by the Fluor Daniel Lead Structural Engineer.

F. Areas subject to chemical exposure, as designated on the design drawings, shall have a minimum cover of 2 inches, unless otherwise noted on the design drawings.

G. Install and splice waterstops according to the manufacturer's recommendations.

H. Embedments shall be installed to the following tolerances unless noted otherwise on the design drawings:

1. Anchor Bolt Projection: + 1/4 inch, -0.00".

2. Center to Center of any two anchor bolts within a bolt group: + or - 1/8 inch, where a bolt group is defined as the set of anchor bolts for a single fabricated steel shipping piece, or a single piece of equipment or skid.

3. Center to Center between bolt groups: + or - 1/4 inch.

4. Anchor bolt Plumbness: 1/8 inch in 3 feet.

5. Plate Inserts: + or - 1/4 inch horizontal and + or - 1/32 inch vertical

Note!!! The Fluor Daniel Lead Structural Engineer shall verify and delete the above listed tolerance requirement if specified on the project Standard Drawings titled "Concrete General Notes", "Steel General Notes" and/or "Anchor Bolt, Sleeves and Nut Fabrication and Installation Details".

3.3 Joints

A. Construction, isolation, and control joints are to be built in accordance with and located according to the design drawings. The Fluor Daniel Lead Structural Engineer shall approve any variation from location shown.

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B. Saw cut control joints as soon as the concrete is hard enough to prevent raveling of the surface and dislodging of the aggregate, but no later than 12 hours after concrete placement. Prime and seal joints according to sealant manufacturer's instructions. Control joints are to be cut in strict accordance with the saw manufacturer's written recommendations. Determine the sawing sequence based upon slab pour time and size.

C. Control joints in slab toppings are to be located directly above and in line with the control joints in the underlying concrete slab.

D. Install isolation joints where paving adjoins vertical surfaces such as walls, columns, catch basins, manholes, and equipment foundations. Gap width shall be 1/2 inch.

3.4 Placing Concrete

Note!!! ASTM C94 states that the discharge of ready mixed concrete shall be completed within 1.5 hours of batching. If this criteria needs to be modified, then a statement should be added to this section.

A. Batch tickets shall indicate total water permitted for the batch and water quantity added at the batch plant.

B. The Concrete Contractor shall maintain records of concrete placement including record date, location, quantity, air temperature, field test results, and test samples taken. Concrete delivery tickets shall be maintained with records for ready mixed concrete.

C. Concrete that has achieved initial set or has been contaminated by foreign matter shall not be deposited in the structure. Retemperring or addition of water after concrete is first mixed shall not be allowed.

D. Tolerance for floor slabs shall be in accordance with ACI 117 Section 4.5.7 for a conventional bull floated finish.

E. Finish exterior slabs and equipment bases with a "floated finish" and stairs, steps, ramps, and walks with a "broom finish". Finish interior floor slabs with a "trowel finish".

3.5 Curing and Protection

A. General

1. Freshly deposited concrete shall be protected from premature drying and excessively hot or cold temperatures and shall be maintained with minimal moisture loss at a relatively constant temperature for the period of time necessary for the hydration of the cement and proper hardening of the concrete.

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B. Concrete Surfaces in Contact with Forms

1. The time during which concrete surfaces are in contact with wood or metal forms may be considered as curing time. Wood forms shall be maintained in a moist condition until removal. After form removal, the concrete shall be cured until the end of the curing period by one of the methods of concrete surfaces not in contact with forms. Curing time shall commence as soon as the wall forms have been loosened and sprinkling has begun. Wall forms shall be loosened between 24 and 48 hours after concrete placement and sprinkling begins. Wood forms shall be kept moist until the forms are loosened and the curing procedure begins.

C. Curing Hydraulic Structures

1. Hydraulic structures shall only be cured by a wet cure procedure. Moist wood forms in contact with concrete shall not be considered as curing for hydraulic structures. This includes the use of ponding or sprinkling or a moisture retaining fabric. Curing shall occur for a minimum of 14 days. Wall forms are to be loosened and water continually sprinkled between the wall and forms. A structure designated as a hydraulic structure shall be so noted on the design drawings.

D. Sealing and Dustproofing

1. Exposed concrete floor surfaces, as designated on the design drawings, shall be sealed and dustproofed. Where the concrete is cured by using a liquid membrane curing compound, this sealing and dustproofing can be a part of the curing process provided a suitable compound is used. Where some other curing method is used, the concrete surface shall be coated with a liquid sealing and dustproofing compound. Apply compounds in accordance with the manufacturer's instructions.

2. Liquid membrane curing compounds that also seal and dustproof may be used on exposed concrete floors. Do not use membrane forming compounds on surfaces to receive bonded treatments, tiles, adhered finishes, paint, epoxy toppings, tiles, and additional concrete. Unless otherwise noted, hydraulic structures are not to receive sealing or dustproofing agents.

E. Concrete Surfaces Not in Contact with Forms

1. Concrete surfaces not in contact with forms may utilize any of the methods indicated in ACI 301, Section 5.3.6.4 for preservation of moisture, except do not use ponding, sprayed water, or wet sand on interior concrete slabs.

F. Form Removal

1. Wall and soil supported member forms may be removed after 48 hours provided the concrete is sufficiently hard to prevent damage by form removal, and provided curing operations start immediately. Self-supporting member forms may be removed

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after 7 days provided the concrete strength is 80 percent of the 28-day strength. No superimposed load shall be applied before the 28-day strength has been verified by field cured cylinders.

G. Repair of Surface Defects

1. Unless otherwise specified or permitted by the Fluor Daniel Lead Structural Engineer, tie holes, honeycombs, and other concrete surface defects shall be repaired as soon as practicable after form removal at such times and in such manner as shall not delay, interfere with, or impair the proper curing of the fresh concrete. The Fluor Daniel Lead Structural Engineer shall be notified before proceeding with repair if the defect is greater than 3 inches deep and larger than 200 square inches in surface area, or if the depth is over 1/3 the thickness of the member and greater than 6 inches in any other direction.

2. Prepackaged grouts and patching compounds or a patching mortar similar to the concrete mix minus the coarse aggregate may be used with approval from the Fluor Daniel Lead Structural Engineer. Match the color of the surrounding area.

3. Out-of-tolerance slabs shall be repaired by grinding down high points and/or raising low points by using a specified underlayment compound or repair topping if the areas are exposed.

4. Critical slab areas, as identified on the design drawings, must be replaced if out-of-tolerance. In this situation, the Contractor shall submit a demolition and replacement plan to the Fluor Daniel Lead Structural Engineer for review and approval prior to proceeding.

4.0 ATTACHMENTS

Attachment 01: (27Oct99)
Supplier Data Requirements

End of Specification

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Supplier Data Requirements

	Type of Submittal		When Required				Remarks
	For Approval	For Record	Weekly	After notice to proceed	Prior to fabrication	With RFQ Submittal	
Contractor's Q.A. Program	X					X	
Batch Plant & Truck Mixer Certification		X				X	
Mix Designs	X					X	
Trial Batch Qualification Test Results	X					X	
Cement Certifications		X				X	
Mineral Additive Certifications		X				X	If applicable
Fine & Coarse Aggregate Certifications		X				X	
Admixture Certifications		X				X	
Material Suppliers, Sources, and Certifications		X				X	
Manufacturer Spec's, certifications and Installation Instructions		X				X	
Proposed Curing Methods	X					X	
Reinforcing Bending Schedule and Placing Drawings	X				7 days		
Certified Mill Test Reports for each bar size and heat number		X		7 days			If requested.
Epoxy Coating Inspection Reports		X	X				If applicable.
Delivery Ticket		X					With delivery.
Compression Test Reports		X	X				
Compression Breaks below 500 psi required		X					Same day submittal.
Water Testing Reports		X		10 days			If applicable